

Licensed Electrician's Theory (LET) Licence Assessment Sample Paper Three (2020)



Candidate Surname	
Candidate Given Names	

Reference Material

- AS/NZS 3000:2018 Wiring Rules
- AS/NZS 3012:2019 Electrical installations – Demolition and Construction sites
- AS/NZS 3008.1.1:2017 Electrical installations – Selection of cables
- Electricity Safety (General) Regulations 2019
- AS/NZS 4836:2011 Safe working on or near low voltage electrical installations and equipment

Instructions

- Personal notepads and paper are not permitted.
- Pens only must be used. Answers in pencil may not be marked.
- Do not remove any sheets from this assessment paper or the room.
- Papers with no name or signature will not be marked.
- Units must be shown to obtain full marks.

Results

Candidates need to obtain 75% or more to pass this assessment. If a mark of 74% or less is achieved, a minimum of 14 days is required before you are permitted to re attempt the assessment.

I, the above named candidate confirm:

- I understand the instructions provided to me.
- do not have any unauthorised materials in my possession
- have not attempted the Licensed Electrician's Theory Assessment at any venue within the past 14 days.

Candidate			
	Print name	Signature	Date

Working Time: 2 hours and 15 minutes

At the end of this time you will be asked to stop.

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	Total
Mark																			

Final Percentage	Pass/Fail

Supervisor			
	Print name	Signature	Date
Assessor			
	Print name	Signature	Date
Reviewed by (If necessary)			
	Print name	Signature	Date

AS/NZS 3000 WIRING RULES

In the following **four** Wiring Rules questions:

- You are required to write the Wiring Rules Clause and/or Table number in the space provided.
- The correct Wiring Rules Clause and Subclause must be given, eg. 3.5.2 (b) (i)

The correct answer to both parts must be given to obtain full marks.

Question 1. State two protection requirements for the connection of the main earthing conductor to the earth electrode.

1.

Wiring Rules Clause Number: _____

2.

Wiring Rules Clause Number: _____

[1 + 1 + 1 + 1 = 4 Marks]

Question 2. State whether or not it is a requirement for a functional switching device to indicate the "off " or "on" position.

Wiring Rules Clause Number: _____

[2 + 2 = 4 Marks]

Question 3. Determine the voltage rating required for the control circuit wiring and equipment of a step-down autotransformer with an input voltage of 230 V, that provides a 20 V secondary output for a control circuit.

Wiring Rules Clause Number: _____

[2 + 2 = 4 Marks]

Question 4. State the nominal voltage and tolerance for low voltage supply systems and electrical installations in Australia.

Wiring Rules Clause Number: _____

[2 + 2 = 4 Marks]

AS/NZS 3012 CONSTRUCTION AND DEMOLITION SITES

In the following **two** AS/NZS 3012 questions:

- You are required to write the Standard's Clause number and/or Table number in the space provided.
- The correct Clause and Sub-clause number must be given. e.g. 2.10.2 (f).

The correct answer to both parts must be given to obtain full marks.

Question 5. . State the requirements of flexible cord used to provide supply to a false-car working platform.

Standard Clause Number: _____ [2 + 2 = 4 Marks]

Question 6. State one permitted use for socket outlets installed outside of a transportable structure.

Standard Clause Number: _____ [2 + 2 = 4 Marks]

ELECTRICITY SAFETY (GENERAL) REGULATIONS 2019

In the following Regulation question, you are required to:

- write your answers on the line/s below each question
- write the complete Regulation and Sub-Regulation number, if applicable, in the space provided, e.g. 401(e)(3).

The correct answer to both parts must be given to obtain full marks.

Question 7. State the margin of error for the route of an underground electrical line that must be recorded as accurately as practicable.

Regulation Number: _____ [2 + 2 = 4 Marks]

ELECTRIC SHOCK SURVIVAL

Electric Shock Survival

DANGERS

Check for your own safety and the safety of the casualty and bystanders.

HIGH VOLTAGE-Wait until the power is turned off.

LOW VOLTAGE-immediately switch off the power. If this is not practicable, pull or push the casualty clear of the electrical contact using material, such as wood, rope, clothing, plastic or rubber. Do not use metal or anything moist.

RESPONSIVENESS

Check for response (verbal and tactile stimuli), touch and talk.

SEND/Shout FOR HELP

Send a bystander to DIAL 000 Ambulance

If available send for Automatic External Defibrillator (AED)

If alone shout for help.

AIRWAY Place the casualty on his/her back.

Tilt the head back and raise the chin forward.

BREATHING Check for normal breathing, observe chest movement, listen and feel for breathing.

Give two initial breaths.

In the absence of normal breathing, if no one has gone for help, place casualty in recovery position and go for help.

CIRCULATION

Position hands on centre of the chest.

Give 30 chest compressions followed by 2 breaths. Depress breastbone 1 / 3 the chest depth (approx. 4cm or 5cm) at the rate of 100 compressions a minute.

As soon as available attach AED and follow its instructions.

Continue CPR" 30 compressions: 2 breaths.

When casualty is normal breathing returns cease resuscitation and move the casualty into the **recovery** or **coma** position.

Keep a constant watch on the casualty, to ensure that they do not stop breathing again, until trained assistants take over.

Question 8. A person has received an electric shock, and is unresponsive. Power has been turned off and the area is safe. Give two examples of what should you do next?

1.

.....

2.

.....

.....

[2 + 2 = 4 Marks]

CABLE SELECTION**Question 9.**

Two 50 mm² three core X90 insulated, sheathed and armoured copper cables, including earthing conductors, are connected in parallel to supply a three phase load.

The cables are protected by a circuit breaker and installed touching directly in the ground at a depth of 0.5m.

- (i) Neglecting voltage drop, what is the TOTAL maximum current carrying capacity of the cables which form the circuit?
- (ii) What would be the current carrying capacity if the soil resistivity was 1.0°C.m/W?

All calculations including the final answer must be completed to a maximum of **two decimal places**.

Table details and units must be shown below to obtain full marks. Part (ii) is over the page.

Part (i)

	Answer		Answer
Table 3 (____?)		Item	
Table		Column	
Derate/rating table		Column	
Derate/rating table		Column	

Part (i) Answer:

Question 9. Part (ii)

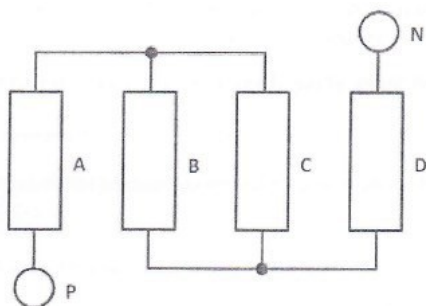
	Answer		Answer		Answer
Derate/rating table		Column		Factor	

Part (ii) Answer:

[1 + 2 + 1 + 1 + 2 + 1 = 8 Marks]

DC CIRCUITS

Question 10.



The following values apply to the diagram above:

The total voltage across points P and N = 160V D.C.

A is 5Ω B is 40Ω C is 10Ω and D is 27Ω

Calculate:

- (i) the current flowing through resistor C
- (ii) the total voltage across resistor B
- (iii) the total power dissipated in resistor D

All calculations including the final answer must be completed to a maximum of **two decimal places**.

(i) Current:		(ii) Voltage:		(iii) Power:	
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MAXIMUM DEMAND**Question 11.**

Calculate the Maximum Demand of a 230V sub-main supplying a distribution board in a non-air conditioned factory installation.

The load connected to the switchboard is:-

- * 3 circuits each of 10 - 10A socket outlets
- * 1 circuit of 2 - 15A socket outlets
- * 2 circuits each of 12 - incandescent lighting points
- * 1 circuit supplying an air compressor motor (12.5 Amps)

All calculations including the final answer must be completed to a maximum of **two decimal places**.

All relevant table details, including table, column and load groups used.

Calculations and units must be shown to obtain full marks.

Table		Column	
Equipment	Load Group	Calculation	Maximum Demand
Total Maximum Demand:			

[1 + 1 + 1 + 1 + 1 + 2 + 1 = 8 Marks]

VOLTAGE DROP**Question 12.**

In a 400V, three-phase, industrial installation, a 30A three-phase drying oven which operates continuously is supplied from a subcircuit originating at a distribution board.

The Consumers mains cables are:

- X90 single-core insulated and sheathed cables with circular aluminium conductors laid in trefoil
- protected by circuit breakers
- not installed with other cables

The submain and final sub-circuit cables are:

- X90 multi-core insulated and sheathed cables with circular aluminium conductors
- protected by circuit breakers
- not installed with other cables

The circuit details are:

Consumer Mains - Single Double Insulated

MD current	65A
Length	120m
Size	35mm ²

Sub-mains - multi-core cable

MD current	40A
Length	42m
Size	25mm ²

Final sub-circuit - multi-core cable

Length	27m
Size	16mm ²

The Consumer Mains and Sub-mains are operating at their maximum permissible operating temperatures. The final sub-circuit cables have an operating temperature of 75°C.

Calculate the total voltage drop from the point of supply to the oven terminals.

All calculations including the final answer must be completed to a maximum of **two decimal places**.

All relevant table details, calculations and units must be shown to obtain full marks.

Cable	Table	Column	Vc	Calculation	Vd
Consumer Mains					
Sub-mains					
Final Sub-circuit					

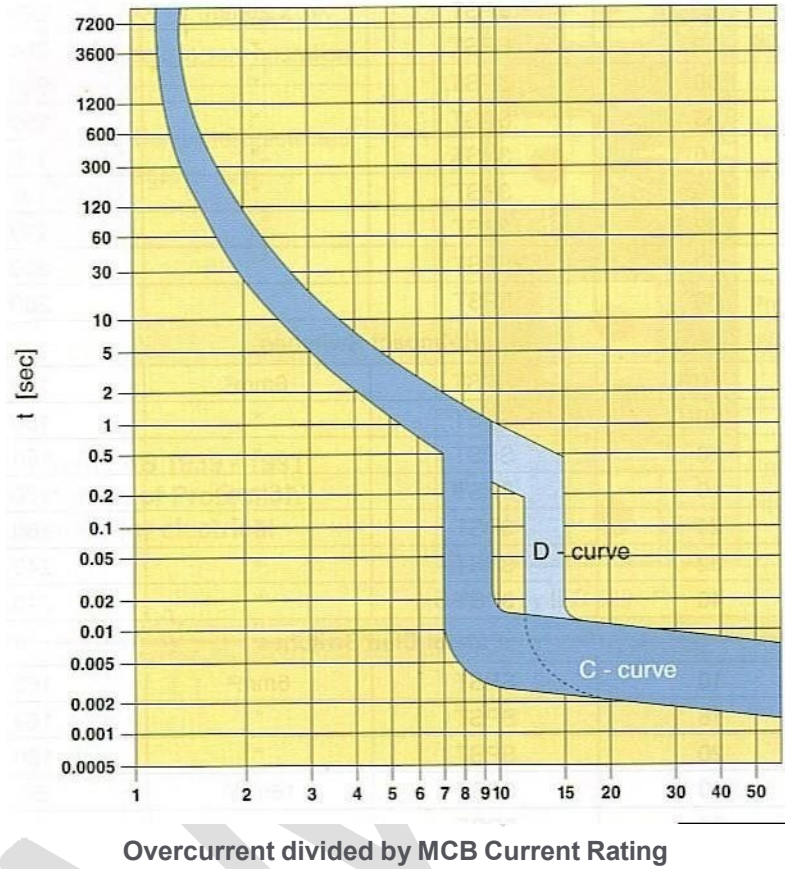
Answer Total Vd:

[1+1+1+1+1+1+1+1+1+1 = 10 Marks]

OVERLOAD AND SHORT CIRCUIT CALCULATIONS

Question 13.

What are the minimum and maximum tripping times for a 60A Type C miniature over-current circuit breaker which is subjected to an over-current of 120A?



Overcurrent divided by MCB current rating:			
Minimum time:		Maximum time:	

[1 + 1 + 1 = 3 Marks]

OVERLOAD AND SHORT CIRCUIT CALCULATIONS**Question 14.**

The main switchboard of a 400/230V industrial installation is directly supplied from a 500KVA transformer which has a prospective fault current of 13,500A per phase.

Submains supply a distribution board from the main switchboard.

The following information is known:

Impedance of the Consumer Mains = 0.00470Ω

Impedance of the Sub-mains cables = 0.02300Ω

Determine the prospective fault current at:

- (i) the main switchboard; and
- (ii) the distribution board.

Work impedances to 5 decimal places.

All calculations must be shown to obtain full marks.

Transformer Impedance:	
MainSw/Bd:	
Dist/Bd:	

[(2+1) + (2+1) + (2+1) = 9 Marks]

RESIDUAL CURRENT DEVICES**Question 15.**

A 30mA Residual Current Device is to be installed in a single-phase sub-main supplying a distribution board.

- * The calculated maximum demand of the sub-main is 16A
- * There are three sub-circuits supplied from the distribution board
- * The circuits are protected by an 8A, a 16A and a 20A circuit breakers

State the minimum current rating of the Residual Current Device.

Wiring Rules Clause Number: _____

[1 + 2 = 3 Marks]

MOTORS AND STARTERS**Question 16.**

CIRCLE the letter in front of the statement you consider to be the most correct.

A secondary resistance motor starter can only be used on a:-

- A. universal motor
- B. squirrel cage induction motor
- C. DC compound motor
- D. wound rotor motor

[2 Marks]

AS/NZS 4836:2011**Question 17.**

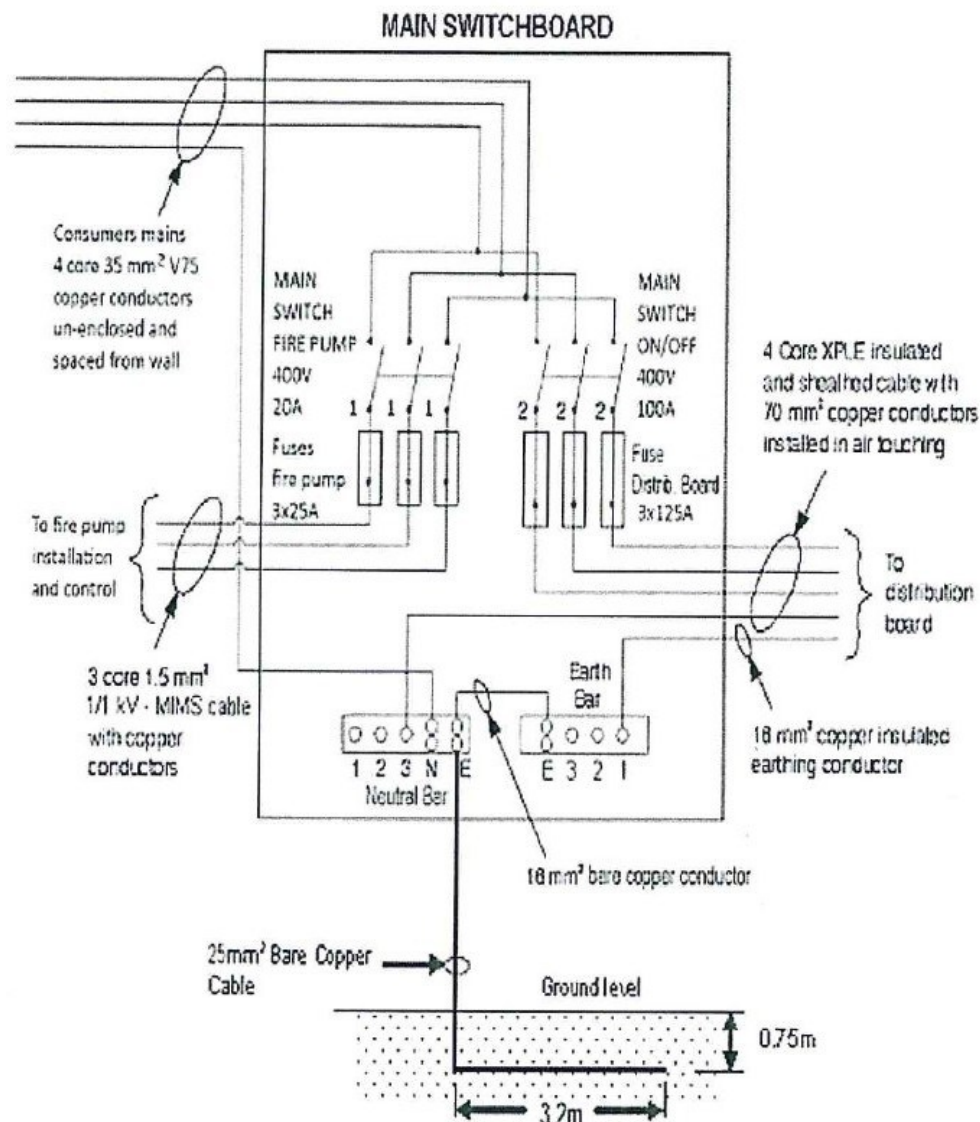
This question relates to AS/NZS 4836:2011.

When should insulated covers and mats be inspected for defects?

Standard Clause Number: _____

[2 + 2 = 4 Marks]

INSTALLATION DEFECTS - NON DOMESTIC



Question 18.

The drawing above shows the MAIN SWITCHBOARD of an industrial installation originating at the consumers mains and contains contraventions to the Wiring Rules.

It supplies a distribution board having a connected load with a maximum demand of 110A per phase and an automatically controlled 3 phase fire pump motor having a current rating of 20A per phase. The fire pump does not require an RCD.

The multi-core MIMS cables are installed spaced from the wall.

Assume the MIMS cables are earthed in accordance with the Wiring Rules and are capable of maintaining supply to the equipment even when exposed to fire and mechanical damage.

All fuses are HRC type. The drawing contains contraventions to the Wiring Rules.

Emergency / Safety equipment installed in the Switchboard is separated from other Switchgear by a metal partition.

The Supply Authority has provided Short Circuit Protection of the point of Supply on the Consumers mains.

Complete the table on the following page.

Question 18. Continued.

Use the diagram on the previous page.

List **FIVE different defects** together with the contravened Wiring Rules Clause/Table number in the table provided below.

Note: Only the first five defects will be considered.

DEFECT DETAILS	WIRING RULE CLAUSE/TABLE No.

[5 x (2 + 1) = 15 Marks]

APPENDIX L
ELECTRIC SHOCK SURVIVAL—Australia

A

EMERGENCY !—Electric shock may cause cardiac arrest.

DANGERS

Check for your own safety and the safety of the casualty and bystanders.

HIGH VOLTAGE—Wait until the power is turned off.

LOW VOLTAGE—Immediately switch off the power. If this is not practicable, pull or push the casualty clear of the electrical contact using material, such as wood, rope, clothing, plastic or rubber. Do not use metal or anything moist.

RESPONSIVENESS

Check for response (verbal and tactile stimuli), touch and talk.

SEND/Shout FOR HELP

Send a bystander to DIAL 000 Ambulance

If available send for Automatic External Defibrillator (AED)

If alone shout for help.

AIRWAY Place the casualty on his/her back.

Tilt the head back and raise the chin forward.

BREATHING Check for normal breathing, observe chest movement, listen and feel for breathing.

Give two initial breaths.

In the absence of normal breathing, if no one has gone for help, place casualty in recovery position and go for help.

CIRCULATION

Position hands on centre of the chest.

Give 30 chest compressions followed by 2 breaths. Depress breastbone 1/3 the chest depth (approx 4 cm or 5 cm) at the rate of 100 compressions a minute.

As soon as available attach AED and follow its instructions.

Continue CPR, 30 compressions: 2 breaths.

When casualty's normal breathing returns cease resuscitation and move the casualty into the **recovery** or **coma** position.

Keep a constant watch on the casualty, to ensure that they do not stop breathing again, until trained assistants take over.

This information is provided for guidance only. It is recommended that persons associated with the installation of electrical equipment or repair of electrical installations obtain Australian Resuscitation Council approved training in current resuscitation methods.